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Implementation and Design of Software for Calibrating a SQUID-AFM Microscope

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Public abridged version — some company-specific content has been removed



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DECLARATION ON THE USE OF ARTIFICIAL INTELLIGENCE TOOLS

Several artificial intelligence tools were used during this internship.

The main tool was *Claude Code*, a coding agent developed by Anthropic. As the internship involved a substantial amount of programming, *Claude Code* made it possible to work in languages I was not familiar with, such as JavaScript. It proved particularly helpful for implementing web pages and for fully restructuring a project in order to change its programming language.

A second tool was *Claude*, Anthropic's conversational assistant, of which *Claude Code* is a paid feature. This assistant was used to structure ideas and to carry out bibliographic research. No confidential company data was submitted to these tools during the internship. For the writing of this report, *DeepL Translate* was used to translate specific technical terms and to improve the English phrasing. *Claude* was also useful for finding and correcting L^AT_EX commands, which helped to comply with the formatting requirements and to improve the overall clarity of the report.

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ABBREVIATIONS AND ACRONYMS

Acronym	Meaning
LION	Leiden Institute of Physics
AFM	Atomic Force Microscopy
SQUID	Superconducting Quantum Interference Device
SoT	SQUID-on-Tip
TF	Tuning Fork
A-Probe	Akiyama-Probe
ROI	Region of Interest
ROC	Receiver Operating Characteristic

INTRODUCTION

Quantum computing has moved, over the past decade, from a laboratory subject to an industrial one, supported by large public programmes and by an investment in quantum start-ups that reached \$12.6 billion in 2025 [1]. As manufacturers increase the number of qubits per chip, sensitivity to fabrication defects becomes the dominant constraint on production yield, and the ability to locate a faulty component rather than discard an entire chip turns into an economic question. This need for characterisation tools is the market on which the host company operates.

QuantaMap is a start-up founded in Leiden in 2022, which employs fifteen people and builds instruments for the diagnosis of quantum chips. Its microscope combines a cryogenic atomic force microscope with a quantum sensor fabricated at the apex of the probe, and measures topography, magnetism, electrical current and dissipation in a single scan.

Any such scan begins with the same step: bringing the probe above the region of interest, while avoiding the fragile structures that would damage it. That step was performed manually by the technicians, by eye, through two lateral cameras. The objective of this internship was to replace it with an automated, repeatable procedure, applicable to an arbitrary chip geometry, establishing the correspondence between a top-view image of the chip and the mechanical reference frame of the positioning stage.

At the start of the internship, no calibration existed. The technicians knew approximately where the region of interest lay on the chip, and displaced the stage so as to bring the tip roughly above it, without any reference frame relating the chip to the stage coordinates. A first attempt at formalising that correspondence was made in the opening weeks, by aligning the tip onto the corners of the chip through a crosshair overlaid on the camera feeds. It yielded a positioning error of approximately $80\text{ }\mu\text{m}$, of the same order as the structures to be reached, and depended on a sample geometry presenting sharp corners. That result led to a redefinition of the project scope.

The retained approach separates the problem into two physical stations. The first acquires a calibrated and annotated reference image of the chip on a dedicated optical bench, where the optical conditions are favourable and a top view is available. The second recovers, once the chip is inserted in the microscope and the chamber closed, the correspondence between that image and the stage frame, using the two lateral cameras alone. All image processing relies on classical computer vision rather than on learning-based methods, the setup being controlled and reproducible and no training data being available. The operator remains the final decision-maker at each validation step.

This report follows that structure. Chapter 1 presents the host company and Chapter 2 the

instrument it builds, with the two technologies it relies on. Chapter 3 covers the first station, from the construction of the optical bench to the software delivered and the metrological investigation it gave rise to. Chapter 4 reports the preliminary work carried out on the second station and the calibration model derived from it.

THE HOST COMPANY: QUANTAMAP

While the theoretical foundations of quantum mechanics date back to the early 20th century, it was not until the 1980s, with the pioneering work of Richard Feynman and later David Deutsch on quantum computing, that the idea emerged of directly harnessing these phenomena as a computational resource, rather than merely as subjects of fundamental study. However, this second quantum revolution has only recently translated into genuine industrial momentum: the launch of the European Quantum Flagship in 2018, the U.S. National Quantum Initiative that same year, and then the quantum component of the France 2030 plan in 2021 marked a turning point by mobilizing massive public funding for these technologies.

The quantum technology industry is now among the sectors experiencing the most sustained growth: investment in quantum technology start-ups reached \$12.6 billion in 2025, 6.3 times the 2024 figure, and the market for quantum hardware, software and services is projected to reach \$60–100 billion worldwide by 2035 [1]. This context is fostering the emergence of a large number of start-ups alongside an ecosystem historically driven by major industrial groups and public research laboratories.

This chapter introduces the company in which the internship took place. It describes its origin and positioning within that ecosystem, its organisation and workforce, the activities and revenue streams through which it operates, and the outlook it sets for itself.

1.1 Introduction to QuantaMap

QuantaMap is a *startup* founded in 2022 in Leiden. It was born out of a specific need identified by its founders: the ability to measure, using a single microscope, multiple parameters such as current, magnetic field, temperature, microwaves, frequency, and even the topography of a sample. Recognizing that this need, already essential to basic research, would become increasingly important with the growth of the quantum computing market, Kaveh Lahabi, Milan Allan, Johannes Jobst, and Jimi de Haan, all researchers at Leiden University, founded QuantaMap.

The company aims to provide *plug-and-play* microscopy systems as well as associated diagnostic services, with the ambition of becoming a leading provider of diagnostic solutions within the quantum ecosystem. This sector now includes major established players such as

IBM and Google, as well as newer companies such as Alice and Bob and IonQ.

This high-potential positioning has quickly enabled Quantamap to establish itself as one of the most promising *start-ups* in the Netherlands. The company received 1.4 million euros in funding from Quantum Delta NL, a Dutch ecosystem dedicated to quantum technologies that aims to generate significant societal impact through technological development.

Quantamap has several partnerships with other players in the sector, including Orange Quantum Systems and Quantware, while continuing to work closely with Leiden University.

1.2 Company Organisation

QuantaMap currently has 15 members, including three of the company's founders, among whom are the current CEO and CTO. The team also includes a System Architect, a Microscope Designer, an Application Scientist, a Nanofab Engineer, an AFM and Electronics Engineer, an AIT Engineer, a Software Engineer, and a Founders Associate. The company works almost constantly with interns in various roles, such as software development and nanofabrication. On a personal level, I was hired by Vlad Tkachuk, an electronics engineer, and Daan Boltje, a systems architect. However, I worked closely with Luigi Mastrodomenico, a microscope designer, and Nick van Nielen, a software engineer, on the technical aspects.

1.3 Activities and Business model

QuantaMap has maintained close ties with the academic community and the field of fundamental research since its founding. From its very inception, the company has benefited from the support of the Leiden Institute of Physics (LION). Even today, Kaveh Lahabi remains a professor and researcher at Leiden University, QuantaMap's Nano-fabrication laboratories are located within the university's physics department, and the company collaborates with doctoral and postdoctoral students.

The initial focus was on a fundamental understanding of physics rather than industrial applications; the two are now closely linked, as the company's recent activity shows. QuantaMap now takes part in several collaborative projects bringing together academic and industrial partners: DeteQT, launched in April 2024, followed by the MKB Hightech project QUACIMI in June 2025, and more recently QORRELATIONS, started in May 2026, which aims to develop scalable test and measurement solutions for superconducting quantum chips. In parallel, the company reached two significant milestones in February 2026: it was awarded funding from the EIC Accelerator, the flagship programme of the European Innovation Council, and published its first paper demonstrating the performance of its microscope. Figure 1.2 summarises these milestones.

QuantaMap's activity is deliberately confined to a single segment of the quantum value chain:

the company does not build quantum computers or quantum chips, but the instruments that measure them. It positions itself as a supplier of metrology and defect-inspection tools for quantum hardware, and structures its commercial offering around three revenue streams: complete scanning AFM-SQUID microscope systems, SQUID-on-tip probes sold separately as consumables, and Diagnostics as a Service, whereby a customer sends a chip and receives a full nanoscale diagnostic report without acquiring any hardware. This last option currently constitutes the main entry point to the market and already serves several customers, in the quantum hardware industry as well as in academic research.

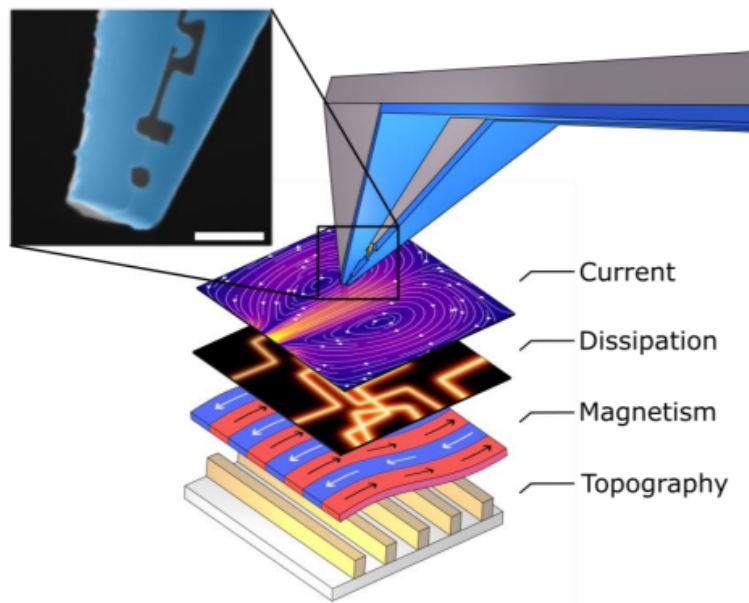


Figure 1.1. Images of the different imaging mode of Quantum Chips by QuantaMap. Reproduced from [2].

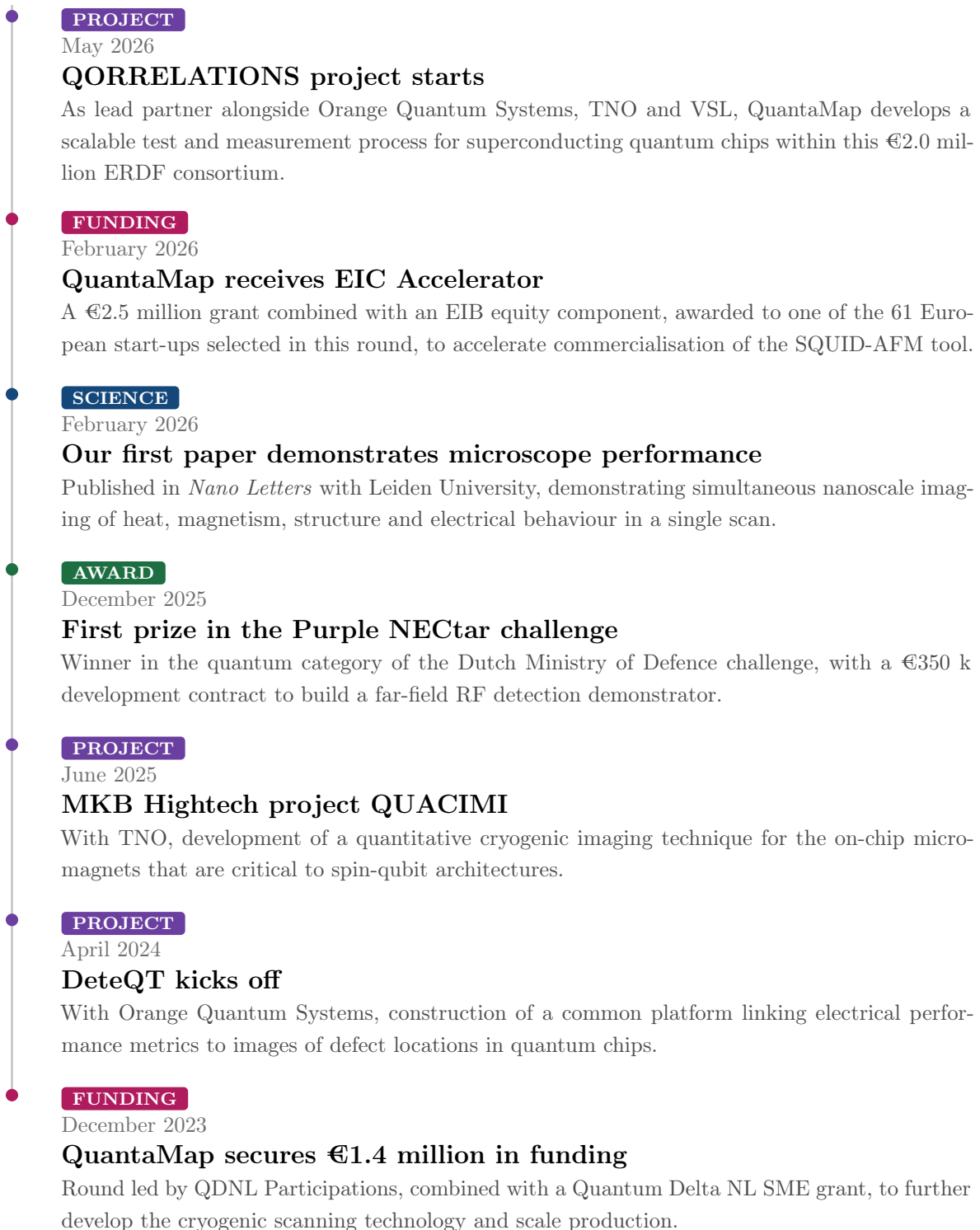


Figure 1.2. Main milestones in QuantaMap’s development, from its initial funding round to its most recent collaborative projects. Source: www.quantamap.eu.

1.4 Perspectives

Demand for the company's systems and diagnostic services is already established, and the market for quantum metrology tools is expected to expand at the same pace as the quantum sector itself. The reason is structural: as manufacturers increase the number of qubits per chip, sensitivity to fabrication defects becomes the dominant constraint on production yield, and the need for dedicated characterisation tools grows with it. The density of the Dutch quantum ecosystem further eases access to this market.

The combination of cryogenic scanning with quantum sensors has, to the company's knowledge, no direct equivalent on the market, and the intellectual property underlying the fabrication process is protected. Other companies address a comparable need with different sensing technologies, whose applicability depends on the type of defect to be resolved.

On this basis, QuantaMap aims to become a reference supplier of diagnostics within the quantum ecosystem, a position it intends to consolidate over the coming years.

THE QUANTAMAP MICROSCOPE

The instrument QuantaMap is building aims to inspect defects in quantum chips, so that a faulty component can be identified rather than the entire chip discarded after a long design and fabrication cycle. It combines a cryogenic atomic force microscope with a patented quantum sensor, a Superconducting Quantum Interference Device fabricated at the apex of the probe.

This chapter presents the two technologies on which the instrument rests, and how their combination produces the measurement the company sells. Atomic force microscopy is introduced first, from its origin to the particular family of self-sensing probes used here; the SQUID is then described, from the Josephson junction upwards; the last section explains what their coupling brings that neither technique provides alone. This background is what makes the positioning problem addressed in Chapter 3 intelligible.

2.1 Atomic Force Microscopy

2.1.1 Historical Background and Development

Since the 17th century, the performance of optical microscopes has improved considerably, owing to technological advances in lens manufacturing, continually pushing back the limits of small-scale observation. An intrinsic limitation constrains this technique: because light is used, diffraction phenomena arise once the size of the observed objects approaches the wavelength of the incident light, making it difficult to observe details smaller than a few hundred nanometers [3]. In order to observe individual atoms, the Scanning Tunneling Microscope (STM) was invented in 1981 by Binnig, Rohrer, Gerber, and Weibel. This invention, awarded the Nobel Prize in 1986, enabled the study of numerous metals and semiconductors at the atomic scale with remarkable spatial resolution. The STM has one structural limitation: electrical conduction of the studied material is required, as the tunneling current flows between a biased tip and the sample. However, early STM experiments showed that whenever the tip-sample distance was small enough to allow a current to flow, significant forces also arose alongside it. The idea of harnessing these forces soon emerged, giving rise to the Atomic Force Microscope (AFM), invented by Binnig in 1986 [4].

2.1.2 Operating principle

The AFM operates by utilizing a very fine tip mounted at the free end of a flexible microcantilever, a tiny beam that can bend and deflect in response to forces. This microcantilever can move in all directions thanks to a piezoelectric tube. As the probe scans the sample surface, it experiences several interaction forces: van der Waals, electrostatic and capillary forces at longer range, and the short-range repulsion described below once contact is approached. These forces deflect the microcantilever, and the deflection is analysed to reconstruct the path of the probe and, from it, the topography of the studied surface.

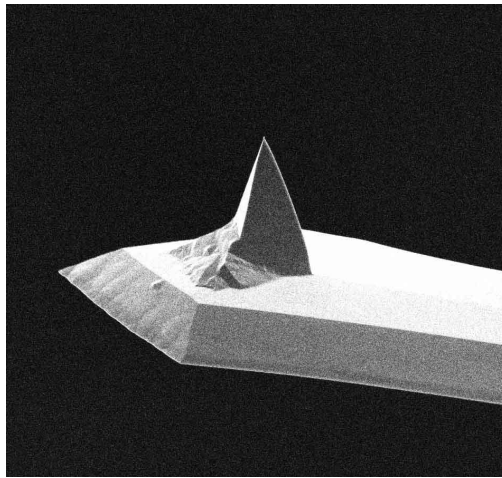


Figure 2.1. Conductive Soft Tapping Mode AFM Probe. Source: www.spmtips.com

The AFM can operate in three primary modes [5]:

- **Contact Mode:** the probe makes direct contact with the surface, and repulsive forces are measured. At such separations the electronic wavefunctions of the tip and sample atoms begin to overlap, producing a strong short-range repulsion arising jointly from the Pauli exclusion principle and from Coulomb repulsion between electrons, which causes the cantilever to deflect.
- **Intermittent Contact Mode (Tapping Mode):** the probe oscillates at its resonance frequency (typically a few hundred kHz). When it interacts with the surface, the oscillation amplitude decreases due to changes in the resonance frequency. This mode is widely used, as it minimizes damage to both the surface and the probe while providing high resolution.
- **Non-Contact Mode:** this mode primarily measures attractive forces between the probe and the surface. Although more challenging to implement, it allows the separation of conservative and dissipative forces and is mainly used under vacuum conditions.

2.1.3 Akiyama-Probe

QuantaMap uses a special subgroup of self-actuating and self-sensing probes based on quartz Tuning Forks (TF) in tapping mode. The TF is used to simultaneously drive and sense a cantilever. A microfabricated U-shaped cantilever with an integrated sharp tip is assembled on a commercially available quartz TF, such that the two legs of the cantilever are symmetrically attached to the two prongs of the TF. This particular type of probe is referred to as the “A-Probe” (for Akiyama-Probe).

What makes this arrangement suitable for the present instrument is that the deflection is read electrically rather than optically. A conventional AFM detects the bending of its cantilever with a laser reflected onto a photodiode, which requires optical access to the probe and an alignment that has to be maintained. Neither is practical inside a closed cryostat, where the probe becomes inaccessible once the chamber is sealed.

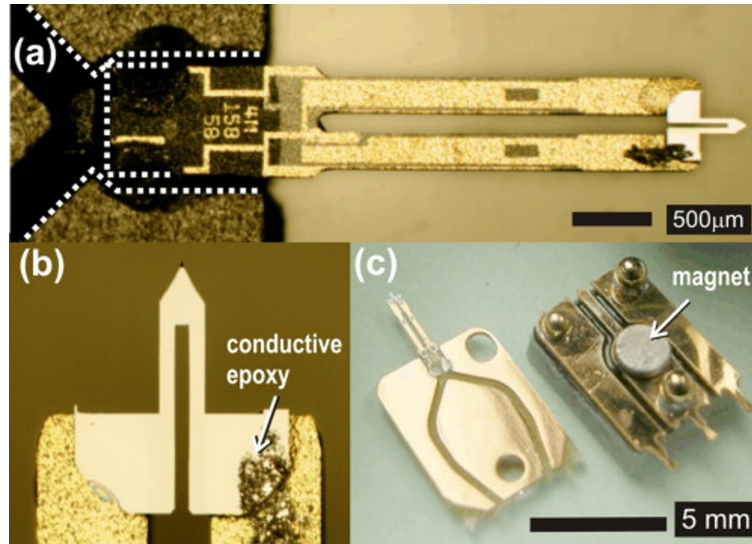


Figure 2.2. (a) Top view of the assembled A-Probe. The dotted lines show the outlines of the metal pieces and the TF, respectively, (b) close up view of the cantilever, and (c) developed probe substrate (left), which can be plugged onto the kinematic mount of receptacle (right) [6].

The U-shaped cantilever and the Tuning Fork are discrete resonators, which can be characterized, in a simplified model, by their respective parameters m_1, k_1, c_1 and m_2, k_2, c_2 . An electrical driving signal is applied to the electrodes of the TF, exciting it at its lowest resonance frequency (Fig. 2.3(c), left). In this mode, the two prongs move in-plane with opposite phases, approaching and withdrawing from each other, with a vibration amplitude on the order of 10 nm. As the cantilever disturbs the symmetry of the TF, a torsional motion of the prongs occurs, resulting in a slight vibration along the z -axis (tip axis) at the glued ends of the cantilever (Fig. 2.3(c), middle). This vibration is amplified by the cantilever, producing a large out-of-plane motion of the tip (Fig. 2.3(c), right).

For practical use, the A-Probe is self-excited (oscillating) at its first resonance peak. When a force is applied to the tip – notably the tip-sample interaction – the resonance frequency is

altered. These variations can be detected electrically and tracked by means of a phase-locked loop (PLL), i.e., a control system generating an output signal whose phase is fixed relative to that of an input signal [6].

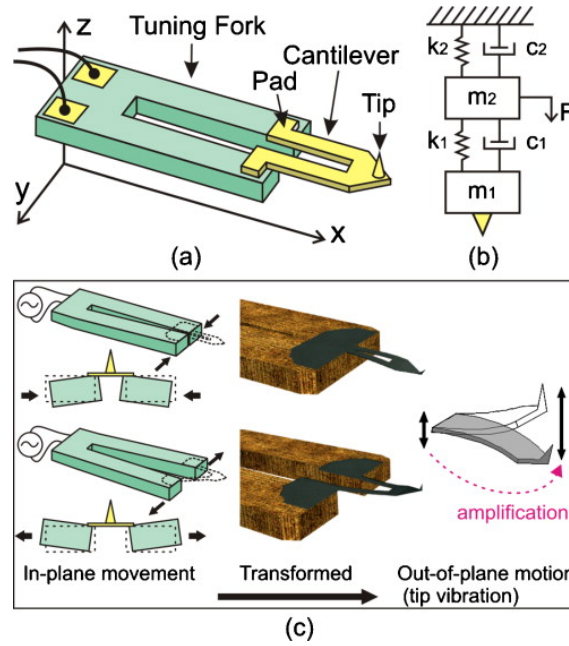


Figure 2.3. (a) Scheme of the standard A-Probe: two discrete resonators, quartz TF, and U-shaped microfabricated cantilever are combined and form an oscillatory force sensor. The piezoelectricity of the quartz TF enables self-sensing and self-actuating of the probe. The spring constant of the A-Probe is determined by the cantilever. (b) A simplified model of the A-Probe. Two resonators are serially coupled. (c) The working principle. In-plane movements of the TF prongs are translated into the vertical motion of the silicon cantilever [6].

2.2 The SQUID technology

2.2.1 Josephson junctions

A Josephson junction, as depicted in Fig. 2.4, consists of two superconductors separated by a thin insulating layer. Superconductivity itself was first observed in 1911 by the Dutch physicist Heike Kamerlingh Onnes during his studies on helium liquefaction: below a critical temperature, the electrical resistance of the material vanishes. In such a junction, Cooper pairs, quasi-particles consisting of electron pairs bound together by attractive interactions typically mediated by phonons in the superconducting material, tunnel through the insulating barrier. This coherent transmission of Cooper pairs, which requires no external voltage, is the Josephson effect.

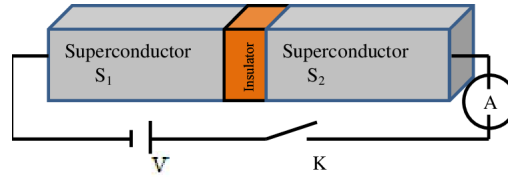


Figure 2.4. Scheme of a SiS Josephson junction. Source: www.researchgate.net

2.2.2 SQUID

The SQUID exploits the Josephson effect to measure extremely small variations in magnetic flux. Typically, a SQUID consists of a ring of superconductor interrupted by one or more Josephson junctions (Fig. 2.5). Superconductors at cryogenic temperatures exhibit surface currents that generate a magnetic field canceling out the external magnetic field. This phenomenon, known as the Meissner effect, expels the magnetic field from the interior of the superconductor and confines it to the surfaces and junctions, concentrating it where it can be most effectively detected. When an external magnetic field is applied to a SQUID, the phase of the Cooper pairs in the Josephson junctions is altered. This phase change alters the critical current of the junctions, so that measuring that current gives access to the magnetic flux threading the ring [7, 8].

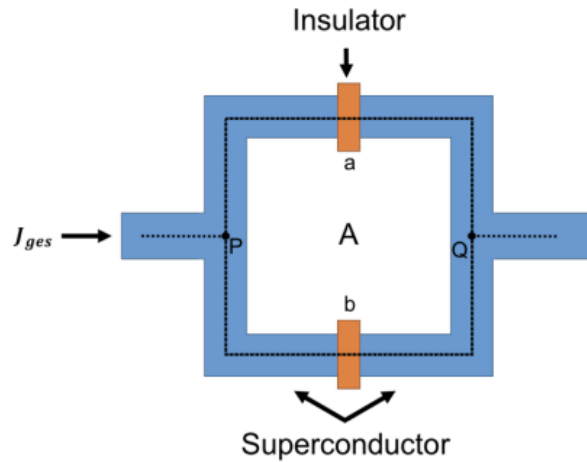


Figure 2.5. Scheme of a SQUID; made up two Josephson junctions a and b connected in parallel [7]

2.3 Cryogenic AFM with a SQUID-on-Tip

QuantaMap's microscope combines the two technologies into a single instrument, compatible with high vacuum, non-magnetic so as not to disturb the SQUID, and able to operate in cryostats at temperatures as low as 100 mK (-273.05°C). The magnetic sensor is a SQUID-on-Tip (SoT), a miniaturised SQUID fabricated at the apex of the probe, shown in Fig. 2.6.

Used on its own, a SoT has no means of knowing its distance to the surface. Both its sensitivity and its spatial resolution depend on that distance, so the sensor must be held close to the sample and at a constant height, which it cannot do by itself. This is what the AFM contributes: the topographic feedback regulates the height continuously, and a single scan therefore returns surface morphology, magnetic field, electrical current and dissipation together, measured at the same points and under a controlled tip-sample distance [9].

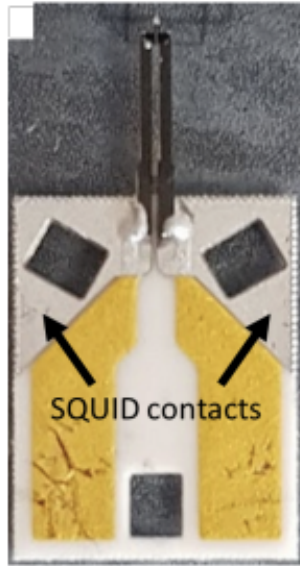


Figure 2.6. Coupling between an AFM and a SQUID

These constraints define the problem addressed in the rest of this report. Once the chamber is closed the sample can no longer be seen from above, the scanner sitting directly over it, and only two lateral cameras remain. The chip is not mechanically constrained on its holder, so its position and orientation differ at every insertion, and the probe itself is mounted by hand and replaced periodically, which displaces the tip within the stage reference frame. Placing that tip over a structure a few tens of micrometres across is therefore not a setting that can be recorded once, but a calibration that has to be redone.

IMPLEMENTATION OF A SCALE CALIBRATION TOOL

Placing the probe over the region of interest is the preliminary step of any scan, and it is currently performed by hand. Replacing that procedure requires two separate pieces of equipment, introduced in Section 3.2.1; this chapter covers the first of them, which produces a calibrated reference image of the chip on a dedicated optical bench.

The problem and the existing setup are stated first, followed by the acceptance criteria against which the work is assessed. The optical bench is then described, then the software built on it, with particular attention to the determination of the scale factor, which proved to be the delicate part. The last two sections report the verification against the criteria and an investigation that occupied a substantial share of the available time, before drawing a critical assessment of the result.

3.1 Context and Problem Statement

3.1.1 The Positioning Problem

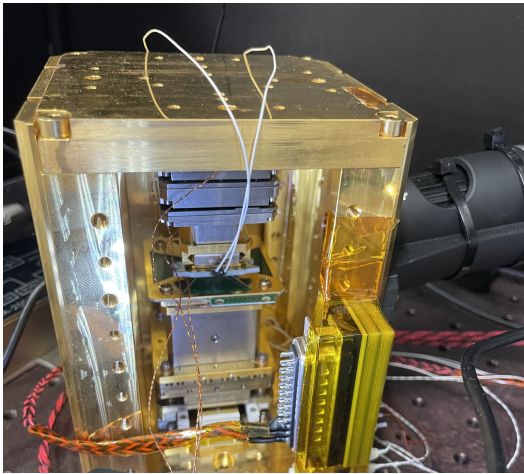
The company's current activities in the diagnosis of quantum chips involve scanning them with an AFM microscope. A fundamental aspect of the AFM scanning process is the positioning of the AFM probe relative to the sample being studied; this is the preliminary step required for a successful scan. In particular, it ensures that the area of interest is scanned rather than another area. In fact, sample analysis generally requires scanning a specific area of the sample while avoiding certain areas of the chip to prevent damage to the AFM tip. This tip localization process is currently performed manually by the company's technicians. An experienced operator does reach the intended area, but by successive visual adjustments: the procedure is slow, and being manual it cannot be standardised, so its outcome depends on who performs it and is not reproducible from one session to the next.

3.1.2 Existing Setup at the Start of the Internship

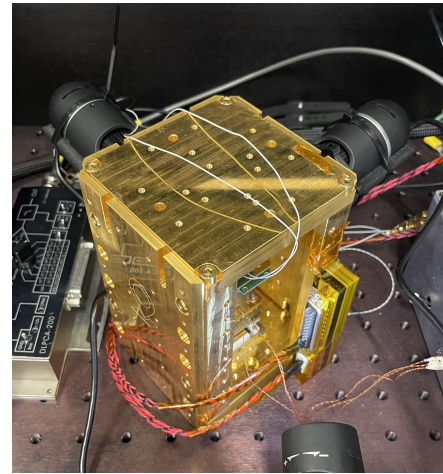
QuantaMap microscopes work with several types of software to control various tasks. The company is now working to consolidate most of this software into a generic platform called NEMO. This platform is a web interface, developed entirely in-house, that brings together five

software programs running on the company's AFM microscopes. For example, the software called R10 controls the AFM system, including tip control, image capture, and the overall scanning process. Another software program called BlueFors fully controls, for example, the AFM's cryostat system. Two additional software programs are implemented in addition to these. Each software program is customized to meet the specific needs of laboratory technicians, making them invaluable tools. As the needs of different technicians evolve, the various features of these software programs must adapt accordingly. So, if new software is developed, it must be compatible with and fully integrable into NEMO.

Regarding the microscope's architecture, the sample scanning station consists of several components. The sample rests on a fixed base, which keeps it completely stationary once it is secured in place. This base is itself mounted on a stage that allows for movement, controlled from a computer workstation, along the three axes of space. These movements can be on the micrometer scale. The tip is then positioned above the sample. When the microscope is not in operation, that is when the cryostat chamber is open, side-view microscopes can be installed to visualize the sample and the AFM tip. The stage's movements allow for coarse repositioning of the sample relative to the fixed tip. The method for positioning the probe involves roughly moving the probe to where the technician believes an area of interest is located. It is therefore a method that, in and of itself, lacks precision and automation.



(a) Front view of the station with the mechanical workbench and one of the two side microscopes



(b) Top view of the closed station with the two side microscopes

Figure 3.1. Image showing the AFM scanning station, which is used to position the sample above the AFM tip. The station includes an automatic stage and side-view microscopes.

3.1.3 Objectives and Scope

This internship project is therefore part of an effort to simplify and improve the accuracy of the procedure for locating the AFM tip. However, the supervisors would like to create a more comprehensive tool that also meets the specific requirements outlined in the specifications later in this report. The main functions the tool must perform are as follows:

- Take a photo of the top of the sample
- Determine the pixel-to-real-distance scale factor for this photo
- Annotate areas of interest to be studied and areas to be avoided to prevent damage to the tip
- Locate the tip relative to the sample
- Enable automatic positioning of the tip on a selected area of the sample

In order to implement this project, I had the freedom to make many technical choices. From an IT perspective, the only requirement was to have a system compatible with the NEMO platform. I was able to work with the hardware and tools the company already had to set up test benches. I was also able to request the purchase of certain hardware to enable more robust implementations at the end of the project.

This chapter presents the work carried out for the first three functions on the list; Chapter 4 covers the last two items.

3.2 Project's specifications

3.2.1 Project breakdown

As previously mentioned in this report, the final results of this project should enable us to fulfill a list of expected functions. The project was subsequently divided into two distinct parts in order to allocate resources consistently based on the equipment required to implement each function. More specifically, the first three expected functions can be implemented using the same physical setup. The last two functions require the microscope itself: its two lateral cameras, its motorised stage and a mounted AFM probe, and therefore access to the cryostat in which it operates. That instrument was needed for their development and testing, and its availability was the main constraint on the project.

Throughout this report, the two stations are referred to by these short, functional names for simplicity, in line with how they are commonly called within the company; their official designations are *Sample Analyser* for Station 1 and *Probe Positioner* for Station 2.

Station 1 is intended to be a standalone pipeline producing a calibrated reference image of the chip. It is built around a dedicated optical bench, physically separate from the AFM system: the chip is mounted on a manual three-axis precision stage, above which a digital microscope is positioned perpendicularly to acquire a high-resolution top view. The pixel-to-millimetre scale factor is obtained by imposing known displacements on the manual stage and measuring the resulting shift in the image, which removes any dependency on a physical calibration target being present in the field of view. The operator then annotates, on this image, the regions of interest to be scanned and the forbidden regions to be avoided, such as

bonding wires and fragile structures. The station outputs a single structured file gathering the calibrated image, its scale factor and the annotated regions. This file is consumed by the second station, which adds the transformation to the stage frame, and ultimately by the navigation interface in NEMO from which the operator selects a target.

Station 2 focuses on establishing the correspondence between that reference image and the physical coordinate system of the AFM. It relies on the existing microscope setup: a closed cage equipped with two lateral digital microscopes, one giving a frontal view and the other a lateral view, and a motorized three-axis stage driven from an algorithm. The essential constraint is that no top view is available once the chamber is closed, since the AFM scanner sits directly above the sample; hence, the need for the reference image acquired beforehand at Station 1. Two quantities must then be recovered from the lateral views alone: the position of the fixed AFM tip in the stage reference frame, and the position and rotation of the chip, which are not mechanically constrained and therefore differ at every insertion. Together, these yield the transformation mapping any point of the top-view image onto the stage coordinates required to bring it under the tip. Figure 3.2 summarizes the various functions of the two stations.

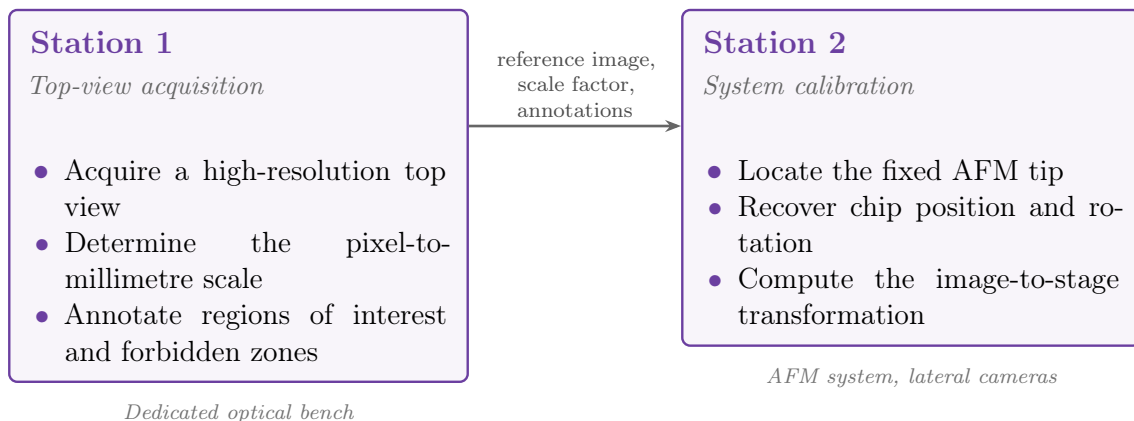


Figure 3.2. Breakdown of the calibration system into two stations

3.2.2 Detailed Specifications

The criteria below concern Station 1 only; those relating to Station 2 are given in Chapter 4. Table 4.1 lists them, distinguishing baseline from stretch requirements.

Two levels of acceptance criteria were defined. Baseline criteria are the minimum requirements for the tool to be considered functional; stretch criteria describe desirable improvements, conditioned on the time available, whose absence does not compromise project completion. Table 3.1 lists the criteria applying to Station 1.

Table 3.1. Acceptance criteria for Station 1, with the associated verification method.

ID	Requirement	Verification
Baseline criteria		
B1	The microscope image can be captured programmatically from Python	Image file produced by a script, with no manual action
B2	The pixel-to-millimetre scale factor is determined to an accuracy consistent with the 25 μm positioning target	Comparison against known displacements of the manual stage
B3	The operator can define and annotate forbidden regions and regions of interest through a graphical interface	Annotations present and correctly located in the output file
B4	The optical setup holds its calibration under normal handling: the scale factor is unaffected by light external contact or by displacement of the bench	Scale factor re-measured after deliberate light contact and after moving the bench
B5	The software is written so that it can be integrated into NEMO, the existing QuantaMap interface, without structural rework	Same framework (NiceGUI) and modular structure; integration reviewed with the development team
B6	The procedure runs without modifying the existing QuantaMap codebase	Standalone execution, no change to the existing repository
B7	The procedure is repeatable across successive chip insertions	Dispersion of the scale factor over several insertions
Stretch criteria		
S1	Chip boundary and orientation are detected automatically, subject to operator validation	Detection compared with manual annotation
S2	The scale calibration step is fully automated	Procedure completed without operator intervention

3.3 Optical Station creation

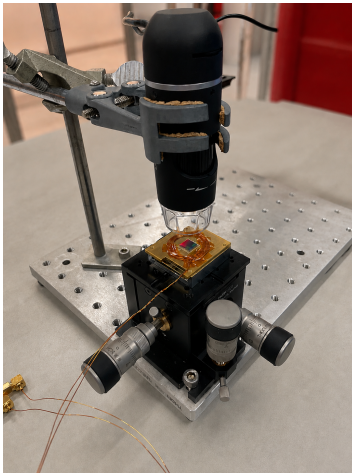
The first step in setting up Station 1 was to build an optical support meeting the requirements defined above. Two successive versions were produced: a rudimentary support, intended to test the first functions as they were implemented, and a consolidated optical station built once the software was complete.

The support is based on a ThorLabs optical breadboard, onto which optical components of the same brand can be screwed, providing a stable and modular base. A ThorLabs manual translation stage, fitted with knobs allowing displacements from a few millimetres down to micrometric increments, is mounted on it and serves primarily as a holder for the quantum chip sample. Top-view imaging is provided by a ToolCraft USB microscope, whose video stream can be visualised live. The microscope is held above the sample by an adjustable metal stand offering two degrees of freedom, for height and for alignment. Figure 3.3a shows this first configuration.

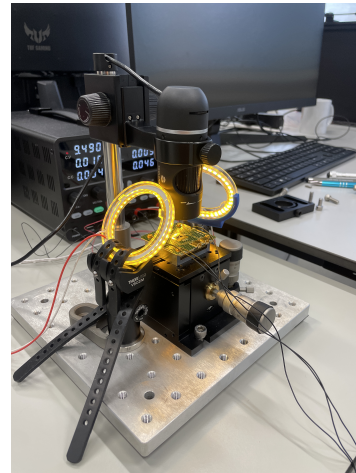
This first setup was the entry point for the implementation of Station 1, described from a software standpoint in the following section. It was used throughout the development of the pipeline and was only improved once the software part had been completed.

This choice was deliberate, but the setup did have several shortcomings. Illumination came solely from the microscope's own light source, which limited image quality. The assembly was also insufficiently rigid, making it liable to move at the slightest external contact, a direct threat to the scale factor and, therefore, to requirement B4.

The station was consequently consolidated, as shown in Figure 3.3b. The adjustable stand was replaced by a fixed microscope mount, insensitive to light contact, and two LED ring lights connected to a current generator were added to provide stronger and more uniform illumination.



(a) First version



(b) Consolidated version

Figure 3.3. The two versions of the optical station.

3.4 Software implementation

3.4.1 Architecture Overview

The software was designed as a set of standalone modules with explicit inputs and outputs, rather than as a single monolithic script. This choice serves two purposes: each step can be tested independently of the others, and the calibration pipeline can be reused without the graphical interface, for instance, from a script or later from the QuantaMap platform itself.

A strict separation is maintained between the algorithmic layer, which performs image acquisition and computes the scale factor, and the interface layer, which only handles user interaction and display. No image processing is carried out in the interface code. Beyond readability, this separation is what makes requirement B5 achievable: the algorithmic layer carries no dependency on any particular interface framework.

The interface was initially written in HTML, CSS and JavaScript, a stack chosen for its familiarity and for the speed at which a first working prototype could be obtained. It was only later that the interface was rewritten in NiceGUI, the framework already used by NEMO, in order to satisfy requirement B5. This rewrite was carried out with the assistance of *Claude Code*, which made it possible to translate the entire interface into a framework I had no prior experience with, within a timeframe that would otherwise have been incompatible with the schedule.

Retrospectively, adopting NiceGUI from the outset would have avoided this rewrite. It was not the right choice at the time: the priority of the first weeks was to obtain a working chain on real hardware quickly, in order to test the acquisition and the scale measurement, and doing so in a framework I had never used would have slowed that down. The separation between the algorithmic and interface layers nonetheless limited its cost: only the interface layer had to be translated, the calibration modules being left untouched. This is, in itself, a practical validation of the architectural choice described above.

The resulting processing chain is shown in Figure 3.4: the microscope stream is acquired, a known displacement of the manual stage is tracked to derive the scale factor, the operator annotates the regions of interest and the forbidden regions, and all this information is finally written to a single output file consumed by Station 2.

3.4.2 Image Acquisition

The microscope is exposed by the operating system as a standard USB video device and is opened directly with OpenCV, with no manufacturer-specific driver. A single capture handle is shared between the live video stream and the still captures, which are the two elements that need access to the device.

Two resolutions coexist. The live preview runs at a reduced resolution, since many drivers

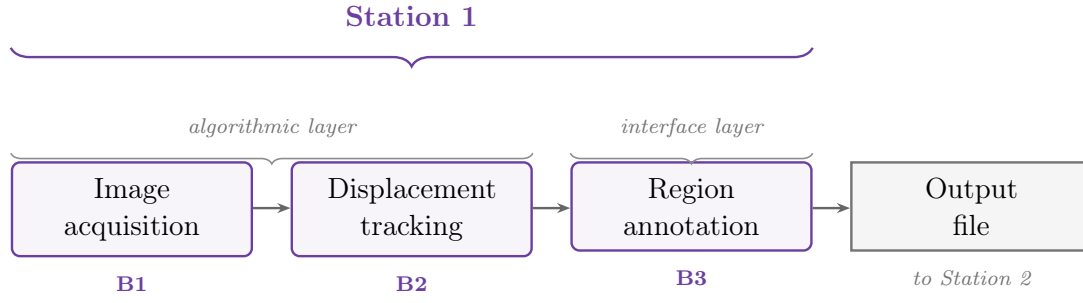


Figure 3.4. Processing chain of the Station 1 software. Each step is implemented as a standalone module with explicit inputs and outputs; the label below each block recalls the requirement it addresses. The output file gathers the calibrated image, the scale factor and the annotated regions, and is the sole input to Station 2.

throttle the frame rate severely at full resolution, which would make the stream unusable for positioning the sample. Everything that contributes to the scale factor, however, is captured at the sensor’s full resolution: halving the resolution would double the relative error on the measured displacement, and the annotation step later applies the scale factor to the chip reference image, so the two must share the same pixel density.

Switching between these two resolutions requires releasing and reopening the device, which resets the driver’s autofocus and exposure. Had this occurred between the two images of a calibration pair, the optical flow used to measure the displacement would fail. This was not a hypothetical concern: the original implementation reopened the camera at every capture, which made a single capture take about thirty seconds and caused the scale computation to fail roughly one time in four. The current implementation keeps the device open throughout a session and only switches resolution at its boundaries (Figure 3.5).

Exposure and focus are then locked manually, after discarding a few warm-up frames, so that no optical parameter can drift between two images of the same measurement.

3.4.3 Determination of the Scale Factor

3.4.3.1 Principle

There is no calibrated target in the field of view, so the scale factor cannot be read directly from the image. It is instead derived from a displacement that is known by construction: the operator turns the stage micrometre screw by an exact number of turns, and the resulting shift is measured in pixels from the two images taken before and after. The scale factor follows from the ratio of the two:

$$\text{px_per_mm} = \frac{d_{\text{px}}}{n_{\text{turns}} \cdot d_{\text{turn}}} \quad (3.1)$$

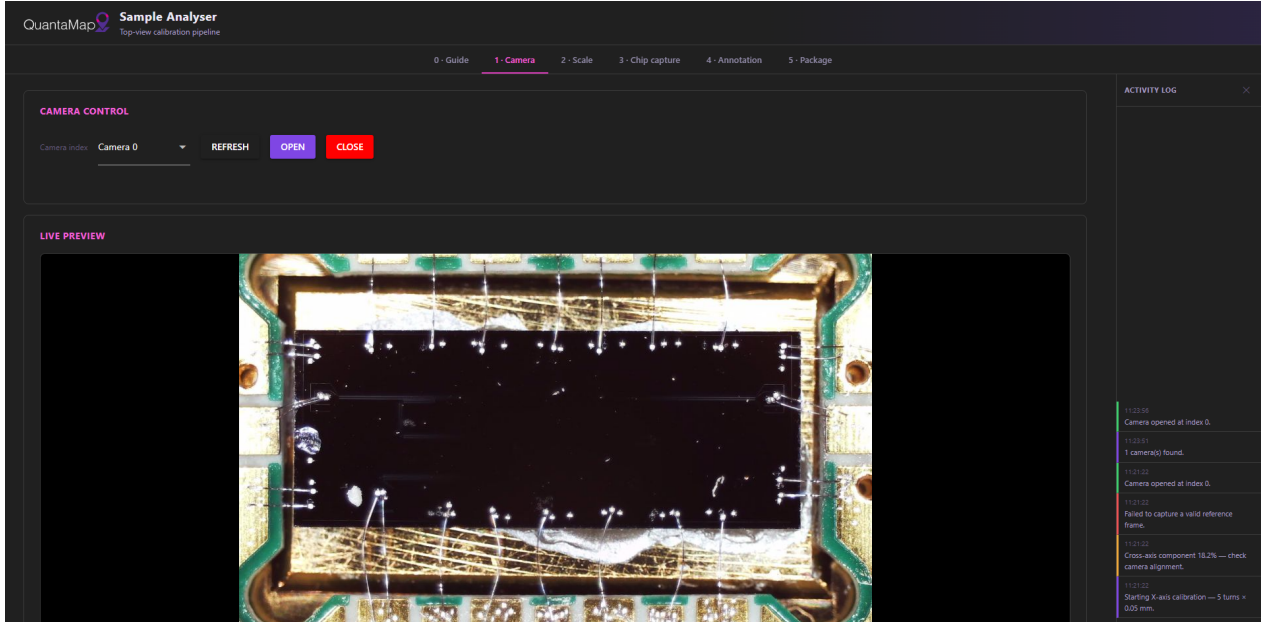


Figure 3.5. The Camera tab during live acquisition, once the camera reopen bug described above had been fixed.

Where d_{px} is the displacement measured in pixels, n_{turns} the number of turns applied, and d_{turn} the nominal displacement per turn given by the stage manufacturer, in millimetres.

3.4.3.2 Displacement Tracking

The displacement between the two images is measured on a small number of points, chosen manually by the operator on distinctive features of the sample. Each point is followed from one image to the other using Lucas–Kanade optical flow [10], which estimates the local displacement of a point by assuming that the intensity pattern in its immediate neighbourhood is preserved between the two images.

This assumption breaks down for large displacements, since a point can move outside the search window before any match is attempted. The pyramidal variant [11] addresses this by first estimating the displacement on a coarsened, downsampled version of the images, then refining that estimate at increasingly finer scales. The coarsest level at which tracking is attempted is controlled by a single parameter, `maxLevel`: at level ℓ , an image is downsampled by a factor 2^ℓ , so a real displacement of d_{px} pixels appears as only $d_{\text{px}}/2^\ell$ pixels at that level. If this reduced value falls outside the search window at the coarsest level used, tracking fails outright rather than degrading gracefully, which turned out to be the dominant cause of unreliable calibrations, discussed in Section 3.5.2. The displacement retained for the scale factor is the median over all tracked points rather than the mean, so that a single mistracked point does not bias the result:

$$d_{\text{px}} = \text{median}_i (d_i) \quad (3.2)$$

where d_i is the displacement measured for point i .

Each point is also checked by tracking it backward, from the second image to the first. The distance between this backward-tracked position and the original point is the forward-backward error:

$$e_{\text{fb}} = \|p_{\text{original}} - p_{\text{backward}}\| \quad (3.3)$$

A point whose error exceeds a fixed threshold is flagged, not discarded: since the operator chose these points explicitly, silently removing one would change what was asked to be measured. This check, however, only catches inconsistent tracking. A point that locks onto a neighbouring instance of a repeated pattern, which quantum chips often exhibit, can track backward through the very same wrong match and therefore pass the check while still being wrong.

3.4.3.3 Computation and Averaging

The scale factor follows directly from the displacement measured in Section 3.4.3.2, using Equation (3.1). The dispersion of the individual point displacements around their median is also reported, as the standard deviation over the axis considered:

$$\sigma_{\text{px/mm}} = \frac{\sigma_{\text{px}}}{n_{\text{turns}} \cdot d_{\text{turn}}} \quad (3.4)$$

where σ_{px} is the standard deviation of the individual point displacements used to compute the median in Equation (3.2).

This quantity must be interpreted with care. It only captures the point-to-point spread of the optical-flow measurement within a single pair of images: it says nothing about the accuracy of the mechanical displacement itself, nor about the reproducibility of the measurement from one calibration to the next. A tight $\sigma_{\text{px/mm}}$ therefore indicates that the tracked points agree with each other, not that the resulting scale factor is correct.

This distinction is not a theoretical caveat. An early estimate of the stage's true displacement per turn, obtained from several calibrations that appeared highly consistent with one another, was later found to be wrong once an unrelated tracking defect was corrected: the apparent agreement between measurements had been produced by the same defect acting consistently

across all of them, not by an accurate measurement. A tight standard deviation is therefore a necessary but insufficient condition for a result to be trusted; establishing accuracy requires comparison against an independent reference, which is addressed in Section 3.5.2.3.

3.4.4 Chip Reference and Annotation of Regions

Once the scale factor is known, a reference image of the sample is captured and cropped to the polygon traced by the operator around the chip boundary. This step performs a crop only, never a resize: rescaling the image would silently change its pixel density and invalidate the scale factor computed in Section 3.4.3, since that factor was derived for the sensor’s native resolution.

The operator then draws on this cropped image, the regions to be scanned and the regions to avoid, such as bonding wires or otherwise fragile structures. Both are stored as pixel coordinates in the cropped image’s own frame, together with the crop offset relative to the original capture, so that they can later be converted to physical positions using the scale factor without ambiguity.

Figure 3.6 shows the result of this process on a real sample, with regions of interest and forbidden zones annotated on the cropped reference image.

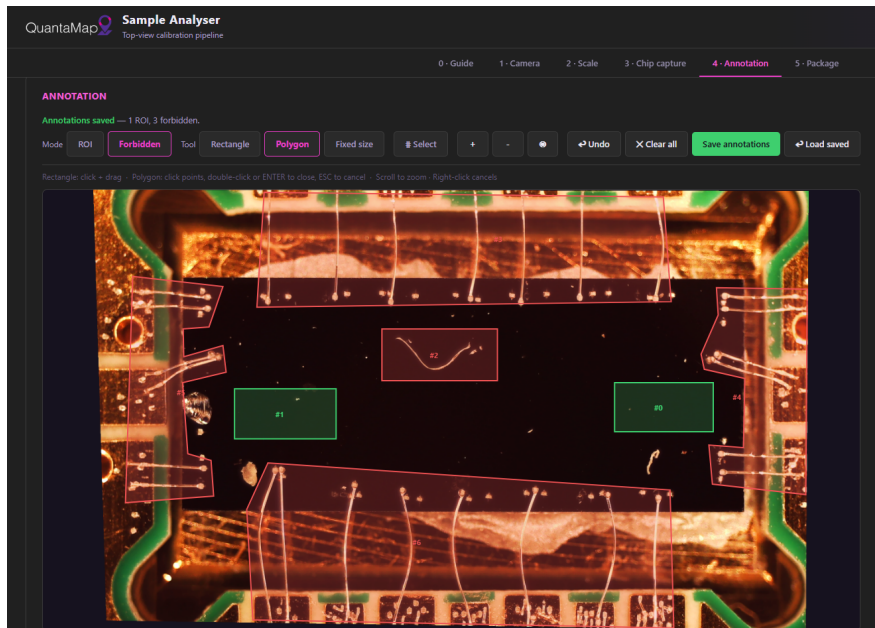


Figure 3.6. Chip reference image after cropping, with regions of interest (green) and forbidden zones (red) annotated by the operator. Image taken from the implemented software.

3.4.5 Output File Format

The output of Station 1 is a single JSON file, consumed by Station 2, containing everything needed to convert a pixel position in the reference image into a physical displacement of the

stage: the scale factor for each axis with its standard deviation, the path and dimensions of the cropped chip reference image, and the complete set of annotated regions.

A value that could not be measured is always written as an explicit `null`, never as a fallback such as zero. This matters because a computed value of zero looks identical to a genuine zero, whereas `null` unambiguously means that no measurement is available. The same convention was applied to the standard deviation discussed in Section 3.4.3.3: writing 0.0 on a failed computation would have suggested a perfectly precise result rather than an absence of one.

For the same reason, the file is only generated once both axes have been calibrated. Generating it from a single axis would force a choice, either omitting the missing axis silently or filling it with an arbitrary value, both of which would misrepresent the state of the calibration to Station 2. Listing 3.1 shows an excerpt of this schema on a real calibration.

```
1 {
2   "scale_calibration": {
3     "px_per_mm_x": 281.5216,
4     "px_per_mm_y": 281.5216,
5     "sigma_px_per_mm_x": 1.1024,
6     "sigma_px_per_mm_y": 0.5501,
7     "pixels_are_square": true,
8     "xy_scale_ratio": 1.0
9   },
10  "annotations": {
11    "roi": [ { "label": "#0", "shape": "fixed", "points": [ ["...", "..."]
12                ] } ],
13    "forbidden": [ { "label": "#1", "shape": "rect", "points": [ ["...", "
14                      ..."] ] } ]
15  }
16 }
```

Listing 3.1. Excerpt of the Station 1 output file, for a sample with one region of interest and three forbidden regions.

3.5 Verification and Results

3.5.1 Verification Against Requirements

Table 3.2 revisits the acceptance criteria defined in Table 3.1, indicating for each whether it was met and by which evidence. The discrepancy investigated in Section 3.5.2 is discussed separately and does not, by itself, invalidate any of these criteria: it concerns the residual agreement between the two axes, not their individual compliance with the positioning target. All baseline criteria directly tied to the software architecture (B1, B3, B5, B6) are met without qualification. B2 is established by the external comparison reported in Section 3.5.2.3, the

Table 3.2. Verification of the Station 1 acceptance criteria defined in Table 3.1.

ID	Requirement	Status	Evidence
B1	Programmatic image capture from Python	Met	Standalone acquisition module, no manual step
B2	Scale factor accurate to 25 μm	Met	Compared against a graduated rule after correction of the tracking depth, on both axes, five repetitions each. The nominal 0,05 mm per turn was found to be in error by 3,8% and replaced by the measured 0,0519 mm (Section 3.5.2.3)
B3	Annotation of regions of interest and forbidden zones through a GUI	Met	Feature implemented and used on real samples (Figure 3.6)
B4	Setup insensitive to light external contact or bench displacement	Not tested as specified	The bench was consolidated and its illumination improved, and the pipeline was validated functionally on it (Section 3.3), but the scale factor was not compared before and after a deliberate perturbation
B5	Integrable into NEMO without structural rework	Met	Interface rewritten in NiceGUI; only the interface layer required translation
B6	No modification of the existing QuantaMap codebase	Met	Standalone application, independent repository
B7	Repeatable across successive chip insertions	Partially met	Five repetitions per axis were carried out during the measurement of Section 3.5.2.3, but only the aggregated result was retained, so no dispersion can be reported
S1	Automatic chip boundary and orientation detection	Not implemented	Left for future work (Section 3.6)
S2	Fully automated scale calibration	Not implemented	Left for future work (Section 3.6)

only one in the project not to rely on the internal consistency of the measurement chain. B4 and B7 are not: the corresponding measurements were either not performed as specified or not retained, and both are discussed in Section 3.5.3. Neither stretch criterion was implemented, both having been deprioritised in favour of the investigation described in Section 3.5.2.

3.5.2 The X/Y Discrepancy

The scale factor is measured independently on each axis, and the two results are not expected to differ: the sensor pixels are square by construction, and the two micrometre screws of the stage carry the same nominal pitch. In practice the two axes never agreed. Table 3.3 lists the ratio $r = \text{px_per_mm}_x / \text{px_per_mm}_y$ obtained over the successive calibration sessions of the project.

Table 3.3. Scale factor measured on each axis across the calibration sessions of the project, with the resulting X/Y ratio. All sessions used $n_{\text{turns}} = 5$ and the nominal $d_{\text{turn}} = 0.05$ mm, before the correction of Section 3.5.2.3. The change in the absolute values from July 7 onwards follows the change of capture resolution described in Section 3.4.2 and affects both axes identically.

Date	px/mm (X)	px/mm (Y)	Ratio	Deviation
2026-06-26	167.58	157.78	1.0621	+6.2%
2026-06-29	134.53	146.15	0.9205	−8.0%
2026-07-02	141.52	145.11	0.9753	−2.5%
2026-07-07	74.48	80.03	0.9306	−6.9%
2026-07-13	71.34	70.90	1.0062	+0.6%

The scatter of that ratio, between 0.6% and 8.0%, is one order of magnitude above the dispersion $\sigma_{\text{px/mm}}$ reported for the same sessions, which stays between 0.27% and 0.75% in relative terms. This is precisely the situation anticipated in Section 3.4.3.3: the tracked points agree closely with one another while the resulting scale factor is wrong. The dispersion cannot arbitrate this discrepancy, and three hypotheses were therefore investigated separately.

3.5.2.1 Hypotheses Tested and Discarded

A rotation between the camera and the stage axes. The first hypothesis was that the camera is not aligned with the stage, so that a displacement along one stage axis projects partly onto both image axes. This is plausible in itself, a transverse component of the order of 7 to 8% being present in every session, but it is insufficient as an explanation. Writing the image-to-stage relation as $\mathbf{p} = S \cdot R(\theta) \cdot \mathbf{m}$, where S is a diagonal scale matrix, $R(\theta)$ a rotation and $\mathbf{m}$ the stage displacement in millimetres, the per-axis measurement retains only the principal component of each displacement. Under an isotropic scale, $s_x = s_y = s$, both axes are scaled by the same factor $s \cos \theta$, and the ratio is

$$r = \frac{s \cos \theta}{s \cos \theta} = 1 \quad (3.5)$$

for any value of θ . A raw ratio different from unity therefore cannot be produced by a rotation alone; it requires a genuine anisotropy, $s_x \neq s_y$. The same algebra shows that combining an anisotropy $a = s_x/s_y$ with a rotation yields a corrected ratio always lying between 1 and a : the angular correction can only move the ratio towards unity, and at $\theta = 45^\circ$ it reports perfect isotropy while the physical anisotropy is unchanged. This is why that correction is exposed as an optional field, left unchecked by default.

An anisotropy of the optical chain. The second hypothesis was that the microscope itself does not resolve the two directions identically. It was tested by repeating the calibration with the camera rotated by 90° relative to the stage: an optical anisotropy is carried by the camera and follows it, whereas a mechanical one stays with the stage. The camera proved close to isotropic, with a self-consistency of 0.6 to 1.4%, well below the observed discrepancy.

An inadequate choice of tracking points. The third hypothesis concerned the operator's choice of points, since a point placed on a repetitive structure may lock onto a neighbouring instance of it, as noted in Section 3.4.3.2. Point placement does have a measurable influence: the discrepancy falls to about 8.9% on non-repetitive details, against 14 to 15% on the periodic grid of the chip. An automatic check was therefore implemented to warn the operator when a point appears unsuitable, and two criteria were tested. Local repetitiveness discriminated in the wrong direction, with an area under the receiver operating characteristic (ROC) curve of 0.316, below the 0.5 expected from chance. Filtering on local contrast raised the fraction of correctly tracked points from 12.7% to 22.3% only. Such a modest gain is not the signature of a misplaced point; it indicates that the points fail for a reason common to all of them. The check was removed, and the investigation redirected towards the tracking algorithm itself.

3.5.2.2 *Insufficient Tracking Depth*

The defect was found in the pyramidal search depth of the optical flow, the `maxLevel` parameter introduced in Section 3.4.3.2. At the coarsest level ℓ , the image is downsampled by 2^ℓ , so a real displacement of d_{px} pixels appears there as $d_{\text{px}}/2^\ell$, and this reduced value must fall inside the search window, that is, within $\pm \text{winSize}/2 = \pm 15$ px:

$$\frac{d_{\text{px}}}{2^{\text{maxLevel}}} \lesssim \frac{\text{winSize}}{2} \quad (3.6)$$

The parameter was set to 3. A displacement of 111 px then appears as 13.9 px against the 15 px limit, formally within bounds but with no margin, and degrading well before it in practice. This is not a marginal configuration: at the default settings of the interface, five turns of 0.05 mm at approximately 305 px/mm produce an expected displacement of about 76 px, already inside the degraded range. The effect was quantified by applying known shifts to a real pair of images and varying only `maxLevel`, all other parameters held fixed (Table 3.4).

Table 3.4. Fraction of tracked points passing the forward–backward check, as a function of the imposed displacement, for two values of `maxLevel`. Same images, same points, all other parameters unchanged.

$ d $ (px)	20	40	60	80	100	111	140
<code>maxLevel</code> = 3	64.5%	63.4%	52.8%	32.1%	13.4%	8.8%	4.5%
<code>maxLevel</code> = 5	64.5%	63.9%	63.9%	63.9%	63.9%	63.9%	56.2%

The two settings are indistinguishable below 40 px. Beyond that, the shallower pyramid collapses, while the deeper one holds at the ceiling imposed by the scene itself, close to 64%, a large part of the field being featureless dark die. On a real acquisition pair reported as failing by the operator, raising the depth to 5 took the proportion of valid points from 12.7% to 73.1%, at the cost of a few additional pyramid levels per point.

The reason this defect went undetected for so long is that it does not degrade gracefully. Beyond the bound of Equation (3.6), tracking does not return a noisier displacement: it returns a wrong one, while the status flag of the algorithm still reports success. This is the mechanism behind the observation already made in Section 3.4.3.3, where several mutually consistent calibrations were later found to be jointly wrong. The defect acted identically on every session, and its consistency was mistaken for accuracy.

Two safeguards accompany the correction. A reliability bound $d_{\text{reliable}} = 5 \cdot 2^{\text{maxLevel}} = 160$ px was derived from Equation (3.6), retaining a factor of 5 rather than the nominal 15 since the measurements above show reliability degrading at roughly a third of the naive limit; the interface compares the expected displacement against it before any capture is taken, rather than after a full capture-and-placement cycle. The forward–backward check was also extended to the operator’s manually placed points, where it had not previously been applied.

3.5.2.3 The Mechanical Displacement per Turn

Correcting the tracking depth removes the dominant defect, but not necessarily the whole discrepancy. Returning to Equation (3.1), the scale factor depends on three quantities: the displacement d_{px} measured in the image, the number of turns n_{turns} applied by the operator, and the displacement per turn d_{turn} . The first has just been addressed and the second is

known exactly. The third was, until then, the only term in the chain never measured: it was read from the manufacturer's specification and applied identically to both axes.

It was measured once the tracking correction was in place. A graduated rule was placed on the manual stage, the scale calibration was run on one axis, and a known 5 mm span read on the rule was compared with the distance derived from the freshly obtained px/mm factor. The operation was repeated five times on each axis. It gives a displacement per turn of 0.0519 mm against the nominal 0.05 mm, an error of 3.8%, and this corrected value is now the default in the delivered software.

The magnitude matters. As Section 3.5.3 establishes, the positioning target requires a relative accuracy between 0.25% and 0.8% depending on the sample, so a 3.8% error is five to fifteen times the admissible bound. It was also invisible to every internal check performed until then: applying identically to both axes, it cancels in the ratio r and leaves the dispersion $\sigma_{\text{px/mm}}$ untouched. Only a comparison against an external reference could reveal it. The corrected value is however specific to the stage used here and must be redetermined for any other.

For the same reason, this measurement does not explain the X/Y discrepancy. A common d_{turn} cancels in the ratio, and only a difference between the two screws could account for it. The per-axis results did differ slightly, but the individual measurements were not retained, so that difference cannot be compared with the dispersion of the measurement itself and nothing can be concluded from it. The hypothesis of a genuine difference of pitch between the two screws therefore remains untested, even though the measurement that would settle it has already been performed once.

3.5.3 Remaining Uncertainty

Three limitations remain on the accuracy of the scale factor. The first concerns what $\sigma_{\text{px/mm}}$ measures: as established in Section 3.4.3.3, it is the point-to-point spread of the optical flow within a single pair of images. It ignores the accuracy of the screw pitch, mechanical play, thermal drift and reproducibility between calibrations, and cannot be used to decide whether an X/Y deviation of a few percent is significant. The second is the absence of a repeatability baseline: five repetitions per axis were carried out during the measurement of Section 3.5.2.3, but only the aggregated result was retained, so the dispersion of the procedure remains unknown. Without it, a deviation of a few percent cannot be compared with the natural scatter of the measurement itself. Repeating that acquisition and keeping the individual values requires no additional hardware and is the most useful measurement left to perform.

The third is structural. Each axis is calibrated separately, and a single measurement carries two equations for three unknowns: the two scale factors and the rotation angle. Solving it requires treating both axes jointly. Grouping the two displacements measured for a common stage travel L into a single matrix,

$$M = \frac{1}{L} \begin{pmatrix} d_x^{(X)} & d_x^{(Y)} \\ d_y^{(X)} & d_y^{(Y)} \end{pmatrix} = U\Sigma V^\top \quad (3.7)$$

the singular value decomposition separates the principal scale factors, carried by Σ , from the camera-to-stage rotation, carried by $UV^\top$. This distinguishes an anisotropy from a misalignment, which independent per-axis corrections cannot do by construction. It was not implemented.

These limitations bear directly on the positioning target. A relative error ϵ produces a position error $\epsilon \cdot L$ over a travel L across the chip, so the required accuracy $\epsilon \leq \Delta/L$ depends on the sample: for the $25\text{ }\mu\text{m}$ target, 0.8% on a 3 mm chip but 0.25% on a 10 mm one. The samples processed during this project span that range. On non-square chips the worst case is set by the longest dimension, and since the chip is not mechanically constrained in the stage frame, the same calibration yields a different worst-case error depending on which stage axis the long side happens to be aligned with. The residual X/Y deviation is therefore not a second-order concern.

3.6 Critical Analysis

Station 1 meets its baseline criteria and produces a usable output file, but the way it reached that state is more instructive than the result itself. The same movement occurred three times: automatic feature detection was replaced by operator-selected points, a single scale value by a value accompanied by a dispersion, and the quality indicators were themselves requalified once it appeared that a tight dispersion had been produced by a defect acting consistently rather than by an accurate measurement. Each step replaced a silent automatism by an instrumented judgement, and the recurring lesson is that a measurement chain must be able to report that it does not know.

The main methodological weakness follows from the same observation. A traceable reference was available throughout the project, requiring no purchase and half an hour of work, but it was only brought in at the very end, once the tracking defect had already been corrected. Until then every intermediate result could only be judged by internal consistency, a criterion that proved deceptive for several weeks. When the comparison was finally made it immediately revealed a 3.8% error on the displacement per turn, which no internal check could have detected. The cost of that measurement was negligible; the cost of deferring it was several weeks of misplaced confidence.

The consequence on scope is direct: both stretch criteria were deprioritised in favour of this investigation. The arbitration appears justified, since automating a procedure whose output cannot yet be trusted would have compounded the problem, but it leaves the tool

more manual than intended. Several weaknesses also remain in the delivered software — the output package is written non-atomically, the application depends on the DirectShow backend and has been tested on Windows only, and no automated test covers the algorithmic layer. None prevents operation on the intended workstation, but each would need addressing before integration into NEMO.

Finally, the project did not reach Station 2. The work carried out towards it is limited to the state of the art and feasibility study presented in Chapter 4, and the coordinate transformation between the reference image and the stage frame remains to be established.

THE PROBE POSITIONER: PRELIMINARY WORK AND CALIBRATION MODEL

The second station is the one that establishes the correspondence between the reference image produced in Chapter 3 and the mechanical frame of the stage. It was not implemented. This chapter reports what was carried out towards it: an early attempt at locating the tip by hand, whose result motivated the redefinition of the project, and the calibration model derived afterwards.

The preliminary work and its outcome are presented first, then the acceptance criteria of the station. The following sections establish why a two-dimensional model cannot describe the setup, how the perspective projection can nonetheless be linearised around the operating point, and what joint two-camera model follows from it. The chapter closes on the state of the art that informed these choices and on what remains to be done.

4.1 Context and Preliminary Work

4.1.1 First Approach: Manual Crosshair Alignment

A first attempt at locating the tip was carried out during the opening weeks of the internship, before the work described in Chapter 3. It relied on the operator alone. A crosshair was overlaid on each of the two lateral camera feeds, and the stage was displaced until a visible corner of the chip coincided with it; the stage position was then read and recorded. Repeating this on several corners provides a set of correspondences between image and stage coordinates, from which an affine transformation can be fitted.

4.1.2 Results and Limitations

The fit itself was consistent, with a residual of 0.029 stage units, but the positioning error measured on validation points reached approximately $80\mu\text{m}$. This is to be compared with the structures the microscope must reach, which are of the order of $100\mu\text{m}$: the error is of the same magnitude as the target itself, and the procedure cannot reliably place the tip on a chosen feature.

Three causes were identified. Aligning a corner onto a crosshair by eye is intrinsically imprecise, and its precision depends on the operator. The result is not reproducible from one session to the next, since nothing fixes the judgement of what constitutes alignment. Finally, the method requires the chip to present sharp, unambiguous corners on both camera views, which restricts it to a particular sample geometry and prevents any generalisation.

4.1.3 Consequence: Redefinition of the Project

These three limitations bear on the method itself rather than on its execution, and no refinement of the manual procedure would remove them. This result motivated the redefinition of the project scope, and in particular the separation into two stations described in Section 3.2.1: the reference image is acquired beforehand under favourable optical conditions, and the lateral cameras are used only for what they alone can provide, namely the correspondence with the stage reference frame.

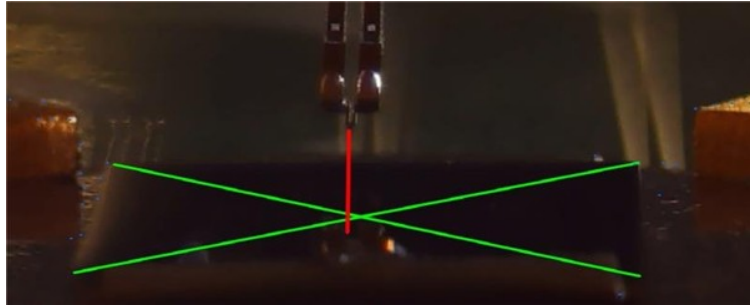


Figure 4.1. Capture of a camera feed during manual alignment

4.2 Specifications

The acceptance criteria for the second station follow the same two levels as those of Section 3.2.2, and are numbered continuously with them. Table 4.1 lists them.

None of these criteria was reached, the station having remained at the stage of the model presented in the following sections. These criteria are demanding, and they need not all be met for the station to be useful. An intermediate mode would already remove most of the manual work: the calibration provides an estimate of the tip position, the stage is moved to it, and the operator corrects the residual offset by eye on the lateral views. Navigation would then start in the immediate neighbourhood of the target rather than from an arbitrary position, which is the expensive part of the current procedure. This fallback was not investigated during the internship, and the criteria of Table 4.1 are stated at their full level, but it is worth recording as a plausible intermediate objective should the full accuracy prove difficult to reach.

Table 4.1. Acceptance criteria for the Probe Positioner, with the associated verification method.

ID	Requirement	Verification
Baseline criteria		
B8	Tip position in the stage frame determined to better than $25\ \mu\text{m}$	Dispersion over repeated determinations
B9	Chip position and rotation recovered from the lateral camera feeds	Comparison with the Station 1 reference image
B10	Complete image-to-stage transformation computed and validated	Residual on known reference points
B11	Positioning accuracy below $50\ \mu\text{m}$, target $25\ \mu\text{m}$	Lateral views used as ground truth
B12	No modification of the existing QuantaMap codebase	Standalone execution, independent repository
B13	Repeatable across successive runs	Dispersion of the transformation over several insertions
Stretch criteria		
S3	Tip localisation and chip registration require no manual alignment	Procedure completed without operator intervention
S4	Automatic rejection and re-run when the validation error exceeds the threshold	Deliberate degradation of a calibration

4.3 The Depth Problem

The difficulty that the manual method concealed is geometric. The tip does not lie in the plane of the chip: it stands at a different height, and the two lateral cameras observe both from an inclined viewpoint. A pixel displacement in either view therefore mixes contributions from all three stage axes, and a purely two-dimensional calibration model cannot separate them.

Each camera is modelled as a pinhole camera, the standard model for an optical system free of major defects. Image formation decomposes into two successive transformations. The extrinsic transformation expresses a point $\mathbf{p} \in \mathbb{R}^3$, given in the mechanical frame of the stage, in the frame of the camera, through a rotation $R \in SO(3)$ and a translation $\mathbf{t}$, both fixed by the physical placement of the camera: $\mathbf{p}_{\text{cam}} = R\mathbf{p} + \mathbf{t}$. The intrinsic transformation then projects $\mathbf{p}_{\text{cam}} = (X, Y, Z)^\top$ onto the sensor plane through the calibration matrix

$$K = \begin{pmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix}$$

where f_x and f_y are the focal lengths in pixels along each sensor axis and (c_x, c_y) the optical centre in the image. Combining both steps, $s\tilde{\mathbf{u}} = K[R \mid \mathbf{t}]\tilde{\mathbf{p}}$ with $s = Z$, which in explicit

non-homogeneous form reads

$$u = f_x \frac{X}{Z} + c_x, \quad v = f_y \frac{Y}{Z} + c_y \quad (4.1)$$

These relations are intrinsically non-linear in $\mathbf{p}$: the pixel coordinate depends on X and Y divided by the depth Z along the optical axis. This dependency in $1/Z$ is what makes each camera sensitive, however weakly, to depth, and it is the physical basis of the model proposed in Section 4.5: two cameras whose optical axes are not parallel provide two independent measurements of that effect, and can in principle constrain the height z of the stage.

4.4 Linearisation: from the Projective to the Affine Model

The joint model proposed below assumes that the observation vector depends affinely on $\mathbf{p}$, rather than through the full perspective projection of Equation (4.1). This assumption can be justified by a first-order Taylor expansion around a nominal depth Z_0 , the working distance of the objective. Writing $Z = Z_0 + \delta Z$ with $\delta Z \ll Z_0$,

$$\frac{1}{Z} = \frac{1}{Z_0 + \delta Z} \approx \frac{1}{Z_0} \left(1 - \frac{\delta Z}{Z_0} \right) + \mathcal{O}\left(\frac{\delta Z^2}{Z_0^2}\right) \quad (4.2)$$

which, substituted into Equation (4.1), makes u a first-order affine function of $(X, Y, \delta Z)$, the error being controlled by the ratio $\delta Z/Z_0$. The affine model is therefore a local linearisation of the perspective projection around the operating point of the setup, not an exact property of the optics. Its validity rests on a condition that is simple to state but must be verified experimentally: the total stage travel covered by the calibration grid, and in particular its extent along z , must remain small compared to the working distance Z_0 .

Three physical effects are not captured by the affine model and may cause the real behaviour to depart from it. The variation of magnification with z is the direct manifestation of the $1/Z$ term neglected beyond first order: the same lateral displacement produces a slightly different pixel displacement depending on the depth at which it is observed. This is the dominant non-linearity to watch, since it grows with the z travel. Depth of field is the second: if the calibration volume exceeds it, points far from the nominal focus plane are blurred, degrading detection precision without necessarily biasing the model. Optical distortion is the third, and differs in nature from the other two, since it depends on the position of the point in the image rather than on its depth; if the calibration grid covers a large portion of the field of view, it may introduce a systematic curvature of the residuals uncorrelated with z , which should not be mistaken for a depth non-linearity.

The affine model is thus a working hypothesis, motivated by the geometry of the setup, and its validity is to be examined a posteriori through the calibration residuals. A systematic structure in the residuals correlated with z , or with cross terms such as xz and yz , would indicate that one of these three effects is no longer negligible and would justify enriching the model rather than keeping its purely affine form.

4.5 Proposed Joint Two-Camera Model

Rather than calibrating each camera separately, both views are treated as a single measurement. Grouping the four pixel coordinates observed simultaneously into one observation vector $\mathbf{o} = (u_1, v_1, u_2, v_2)^\top$, the linearisation of Section 4.4 gives

$$\mathbf{o} = M \tilde{\mathbf{p}}, \quad \tilde{\mathbf{p}} = (x, y, z, 1)^\top \quad (4.3)$$

where M is a 4×4 matrix absorbing, for both cameras at once, the unknown orientations, the focal lengths and the optical centres. No angle needs to be measured and no intrinsic calibration needs to be performed beforehand: all these quantities are estimated jointly from displacements of the stage, which serves as its own ground truth.

The reason for solving both cameras together rather than one after the other lies in the depth. A single inclined camera does constrain z through the $1/Z$ dependency, but too weakly to be usable alone. Two cameras whose optical axes are not parallel provide two independent constraints, and the system becomes solvable. Once M is estimated, localising the tip amounts to inverting Equation (4.3) for the pixel position of the tip measured in both views, which yields its coordinates directly in the stage reference frame, without any alignment procedure.

4.6 State of the Art

Two families of work informed these choices. The joint estimation of the geometry of several fixed cameras observing a common scene is the subject of multi-camera hand-eye calibration, as formulated for industrial robotic workcells [12]. The relevant result there is methodological rather than algorithmic: calibrating each camera separately and combining the results accumulates error, whereas a joint minimisation of the reprojection error over all cameras does not. This is the argument that motivates the single matrix M of Equation (4.3) in preference to two independent per-camera models.

The recovery of the chip position, which remains to be addressed, falls under feature-based image registration. The standard pipeline, detection, description, matching, robust estimation by RANSAC and fitting of an affine model, is reviewed in [13] for images of markedly

different natures. It supports the choice of classical rather than learning-based methods retained throughout the project: heterogeneity between two imaging modalities does not by itself require a learning-based approach.

4.7 Limitations and Next Steps

The work on this station stopped at the model, for want of access to the instrument. Estimating M requires the microscope itself, its two lateral cameras and its motorised stage inside the cryostat in which it operates, and that instrument was not available for the time this work would have taken. The calibration protocol, the estimation of M , its inversion and the experimental validation of the resulting accuracy were not carried out, and none of the criteria of Table 4.1 can therefore be declared met. The chip registration, which requires correlating the lateral views with the reference image produced by the first station, was not addressed at all.

Three points condition the continuation. The validity of the affine hypothesis must be established before anything is built on it, through the residual analysis described in Section 4.4, since an unnoticed departure from linearity would propagate into every subsequent measurement. Feature detection on CAM2 is expected to be the limiting factor, its illumination being markedly weaker than that of CAM1. Finally, and for the reasons set out in Section 3.6, a traceable reference should be established at the outset rather than at the end, so that the accuracy of the station can be assessed against something other than its own internal consistency.

CONCLUSION

The objective of this internship was to replace a manual positioning procedure slow, dependent on the operator and impossible to standardise, by an automated and repeatable calibration establishing the correspondence between a top-view image of the chip and the reference frame of the positioning stage, to within $25\text{ }\mu\text{m}$. A first attempt at formalising that correspondence, by manual crosshair alignment, gave a positioning error of about $80\text{ }\mu\text{m}$ and depended on a particular sample geometry.

That objective was reached in part. The first station is delivered and functional: a web application that captures a top view of the chip, determines its pixel-to-millimetre scale factor from known displacements of a manual stage, lets the operator annotate the regions to scan and the regions to avoid, and packages the whole into a single file. Five of its seven acceptance criteria are met or partially met, the two remaining ones concerning measurements that were not conducted as specified rather than functions that are missing. The second station stopped at the modelling stage: the geometric reason why a two-dimensional calibration cannot describe the setup was established, and a joint two-camera model was derived from a linearisation of the perspective projection, but it was neither estimated nor validated experimentally, the instrument required for that work not having been available. The end-to-end transformation from image to stage coordinates therefore remains to be established, and the target accuracy has not been demonstrated on a complete chain.

The most substantial technical result came from an unexpected direction. A persistent disagreement between the two axes of the scale factor, which internal consistency checks were unable to arbitrate, led to an investigation that occupied a significant share of the schedule. It revealed a defect in the pyramidal depth of the optical flow, which returned wrong displacements while reporting success, and then, through a comparison against a graduated rule, a 3,8% error in the manufacturer's specified displacement per turn. Both were invisible to every check internal to the measurement chain, and both are now corrected in the delivered software. This investigation was prioritised over the optional criteria of the first station, an arbitration that left the tool more manual than intended but avoided automating a measurement whose accuracy was not established.

On the organisational side, the work was conducted autonomously, and the project scope was formally redefined in the seventh week once the preliminary results had shown the initial approach to be insufficient. The software was written as decoupled algorithmic and interface layers, a choice which allowed the interface to be rewritten in the framework used by the company's platform without touching the calibration modules, and which conditions the future integration of the tool.

Several directions follow. In the short term, establishing a repeatability baseline and keeping its individual measurements would close the two outstanding criteria at no hardware cost, and would allow the residual disagreement between the two axes to be attributed or dismissed. Estimating the joint two-camera model, then validating it against the lateral views, is the natural continuation towards a complete calibration. In the longer term, the instrument is intended to operate inside a cryostat, where thermal contraction during cooldown introduces a further displacement of the probe that the present work does not address. More broadly, the difficulty encountered here is not specific to this microscope: any instrument that must position a fixed probe relative to a sample it cannot see directly faces the same question, and the value of an external reference established early, rather than trusted internal consistency, is the transferable lesson of this work.

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ABSTRACT

Quantum chip manufacturers face a growing sensitivity to fabrication defects as the number of qubits per chip increases, which makes the ability to locate a faulty component rather than discard an entire chip an economic question. QuantaMap, a Leiden-based start-up founded in 2022, builds instruments for that purpose, combining a cryogenic atomic force microscope with a quantum sensor fabricated at the apex of the probe.

Every scan begins by bringing the probe over the region of interest while avoiding the structures that would damage it, a step performed manually through two lateral cameras. The objective of this internship was to replace it with an automated and repeatable calibration establishing the correspondence between a top-view image of the chip and the mechanical frame of the positioning stage, to within $25\text{ }\mu\text{m}$. Reproducing the existing manual procedure gave a positioning error of approximately $80\text{ }\mu\text{m}$, of the same order as the structures to be reached, which led to a redefinition of the project into two physical stations.

The first station is delivered and functional. It acquires a calibrated top view of the chip on a dedicated optical bench, determines its pixel-to-millimetre scale factor from known stage displacements tracked by optical flow, and lets the operator annotate the regions to scan and to avoid. A persistent disagreement between its two axes prompted an investigation which revealed a defect in the pyramidal depth of the tracking, returning wrong displacements while reporting success, and then a 3,8 % error in the manufacturer's specified displacement per turn. Both were invisible to any check internal to the measurement chain. The second station stopped at a calibration model: a joint two-camera formulation derived from a linearisation of the perspective projection, not yet estimated experimentally.

RÉSUMÉ

L'augmentation du nombre de qubits par puce rend les fabricants de plus en plus sensibles aux défauts de fabrication, au point que localiser un composant fautif plutôt que rebuter une puce entière devient une question économique. QuantaMap, start-up fondée à Leyde en 2022, construit les instruments correspondants, en associant un microscope à force atomique cryogénique à un capteur quantique fabriqué à l'apex de la pointe.

Toute analyse commence par amener la pointe au-dessus de la zone d'intérêt en évitant les structures qui l'endommageraient, une étape réalisée manuellement au travers de deux caméras latérales. L'objectif de ce stage était de la remplacer par une calibration automatisée et répétable établissant la correspondance entre une image du dessus de la puce et le référentiel mécanique de la platine de positionnement, à $25\text{ }\mu\text{m}$ près. La reproduction de la procédure manuelle existante a donné une erreur de positionnement d'environ $80\text{ }\mu\text{m}$, du même ordre que les structures à atteindre, ce qui a conduit à redéfinir le projet en deux stations physiques. La première station est livrée et fonctionnelle. Elle acquiert une vue calibrée du dessus de la puce sur un banc optique dédié, détermine son facteur d'échelle pixel/millimètre à partir de déplacements connus de la platine suivis par flux optique, et permet à l'opérateur d'annoter les zones à analyser et celles à éviter. Un désaccord persistant entre ses deux axes a motivé une investigation qui a mis au jour un défaut de profondeur pyramidale du suivi, renvoyant des déplacements faux tout en signalant une réussite, puis une erreur de 3,8 % sur le déplacement par tour spécifié par le constructeur, ni l'un ni l'autre détectable par une vérification interne à la chaîne de mesure. La seconde station s'est arrêtée à un modèle de calibration : une formulation bi-caméra conjointe issue d'une linéarisation de la projection perspective.